

UNIVERSITY INSTITUTE OF ENGINEERING

COMPUTER SCIENCE & ENGINEERING

BACHELOR OF ENGINEERING

CSE/ 4th SEMESTER

OPERATING SYSTEM(24CST-205/24ITT-205)

Unit No.3

Chapter No. 3

Lecture No.33

Topic

:File Concepts and Access Methods

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Learning Objectives

- Understand the concept of files, their structure, and their role in data storage within an operating system.
- Analyze different file access methods (sequential, direct, indexed) and their appropriate usage.



File Concept

- OS provides a uniform logical view of data storage
- It abstracts physical storage into a logical unit called a file
- Files are mapped onto physical storage devices (secondary memory)
- A file is a named collection of related data
- It is the smallest unit of logical secondary storage
- Data must be stored within a file

File-Attributes

- A file is named, for the convenience of its human users, and is referred to by its name.
- A name is usually a string of characters, such as example .c.
- When a file is named, it becomes independent of the process, the user, and even the system that created it

File-Attributes

- A file's attributes vary from one operating system to another but typically consist of these:
 - Name
 - Identifier
 - Type
 - Location
 - Size
 - Protection
 - Time, date and user identification

File-Operations

- The OS preforms various basic file operations:
 - Creating a File
 - Writing a File
 - Reading a File
 - Repositioning within a File
 - Deleting a File
 - Truncating a File

File-System Structure

- File structure
- Logical storage unit
- Collection of related information
- File system resides on secondary storage (disks)

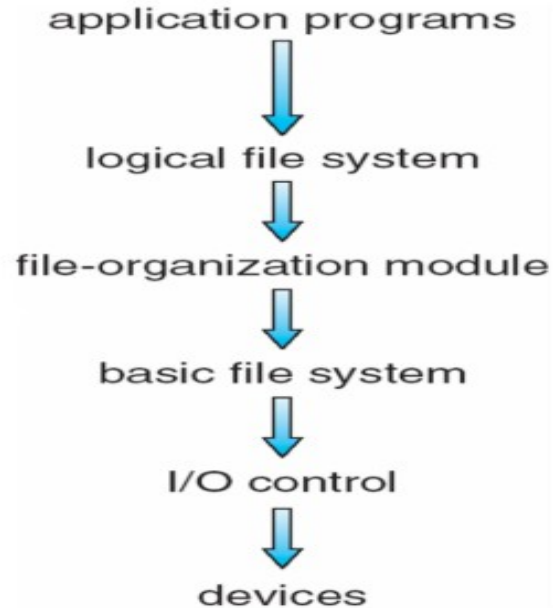
File-System Structure

- Provided user interface to storage, mapping logical to physical
- Provides efficient and convenient access to disk by allowing data to be stored, located retrieved easily
- Disk provides in-place rewrite and random access
- I/O transfers performed in blocks of sectors (usually 512 bytes)

File-System Structure

- File control block – storage structure consisting of information about a file
- Device driver controls the physical device
- File system organized into layers

Layered File Systems



File System Layers

- **Device drivers manage I/O devices at the I/O control layer**
- Given commands like “read drive1, cylinder 72, track 2, sector 10, into memory location 1060” outputs low-level hardware specific commands to hardware controller
- Basic file system given command like “retrieve block 123” translates to device driver

File System Layers

Also manages memory buffers and caches (allocation, freeing, replacement)

- Buffers hold data in transit
- Caches hold frequently used data

File System Layers

File organization module understands files, logical address, and physical blocks

- Translates logical block # to physical block #
- Manages free space, disk allocation



File System Layers

- Logical file system manages metadata information
 - Translates file name into file number, file handle, location by maintaining file control blocks (inodes in Unix)
 - Directory management
 - Protection

File System Layers

Layering useful for reducing complexity and redundancy, but adds overhead and can decrease performance

- Logical layers can be implemented by any coding method according to OS designer

File System Layers

Many file systems, sometimes many within an operating system

- Each with its own format (CD-ROM is ISO 9660; Unix has UFS, FFS; Windows has FAT, FAT32, NTFS as well as floppy, CD, DVD Blu-ray, Linux has more than 40 types, with extended file system ext2 and ext3 leading; plus distributed file systems, etc)
- New ones still arriving – ZFS, GoogleFS, Oracle ASM, FUSE

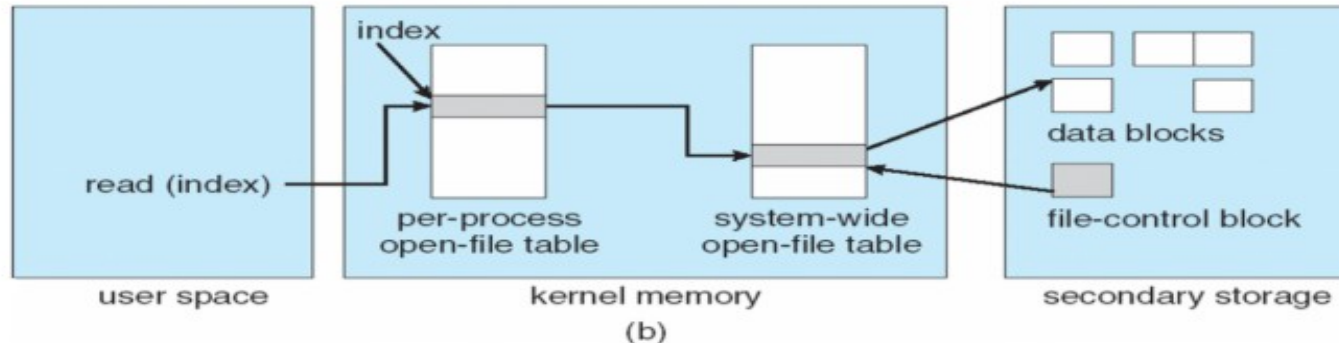
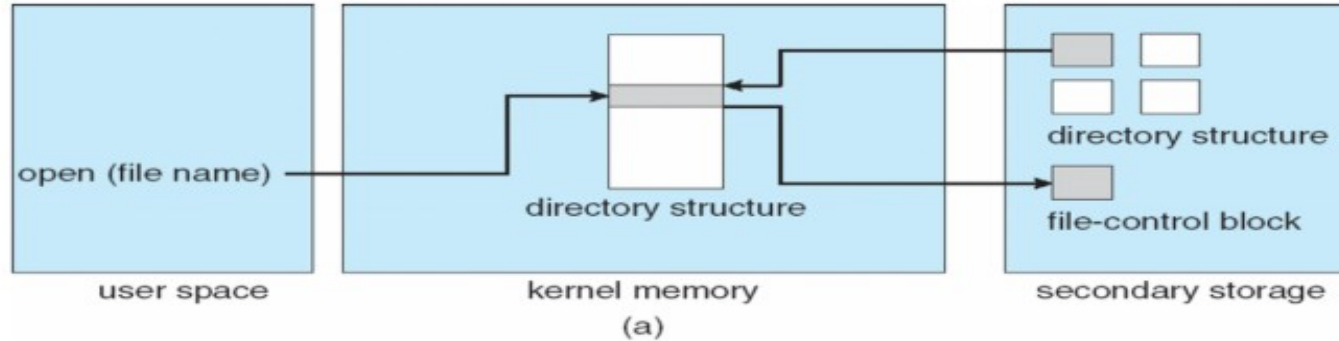
In-Memory File System Structures

- Mount table storing file system mounts, mount points, file system types
- The following figure illustrates the necessary file system structures provided by the operating systems
- Figure 12-3(a) refers to opening a file
- Figure 12-3(b) refers to reading a file

In-Memory File System Structures

- Plus buffers hold data blocks from secondary storage
- Open returns a file handle for subsequent use
- Data from read eventually copied to specified user process memory address

In-Memory File System Structures



Access Methods

- Files store information. When it is used, this information must be accessed and read into computer memory. The information in the file can be accessed in several ways.
 - a) Sequential Access
 - b) Direct Access
 - c) Index Access

Access Methods

a) Sequential Access

- The simplest access method.
- Information in the file is processed in order, one record after the other.
- Reads and writes make up the bulk of the operations on a file.
- A read operation—read next—reads the next portion of the file and automatically advances a file pointer, which tracks the I/O location.
- The write operation—write next—appends to the end of the file and advances to the end of the newly written material (the new end of file).

Access Methods

b) Direct Access

- Also called as relative access.
- A file is made up of fixed length logical records that allow programs to read and write records rapidly in no particular order.
- The direct-access method is based on a disk model of a file, since disks allow random access to any file block.
- For direct access, the file is viewed as a numbered sequence of blocks or records. Thus, we may read block 14, then read block 53, and then write block 7. There are no restrictions on the order of reading or writing for a direct-access file

Access Methods

c) Index Access

- Index contains pointers to the various blocks.
- To find a record in the file, we first search the index and then use the pointer to access the file directly and to find the desired record



Summary of the Lecture

This lecture enables the students to understand the concept of Files are logical units of storage used to organize and manage data in an operating system. Different access methods (sequential, direct, indexed) determine how data is read and written efficiently.



Next Lecture

- **Directory Structure**

Quiz/ FAQ's

Q1. The information in a file is defined by:

- A. Operating system
- B. Compiler
- C. User/creator
- D. Hardware

Q2. Indexed access method uses:

- A. Sequential pointers
- B. Index table
- C. Stack structure
- D. Queue structure

References

<https://www.includehelp.com/c-programming-questions/>

<https://www.studytonight.com/operating-system/>

<https://computing.llnl.gov/tutorials/>

[https://www.tutorialspoint.com/operating_system/index.htm#:~:text=An%20operating%20system%20\(OS\)%20is,software%20in%20a%20computer%20system.](https://www.tutorialspoint.com/operating_system/index.htm#:~:text=An%20operating%20system%20(OS)%20is,software%20in%20a%20computer%20system.)

<https://www.javatpoint.com/os-tutorial>

<https://www.guru99.com/operating-system-tutorial.html>

<https://www.geeksforgeeks.org/operating-systems/>



Faculty-curated videos, NPTEL, Coursera, LinkedIn, or other relevant learning resources

- <https://nptel.ac.in/courses/106105214>
- <https://www.linkedin.com/learning/iot-foundations-operating-systems-fundamentals-2018>



Class-Wise Feedback

Faculty members are encouraged to generate a class specific QR code at the end of each session to collect brief student feedback. This can support self-reflection, improve teaching practices, and allow students to share insights, suggestions, or concerns anonymously.



Thank You

For queries

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